FPGA-based speed adaptive spectral fault-signature detection

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Abstract

Vibration-based fault detection in induction motors relies on identifying characteristic spectral components associated with mechanical and electrical faults. However, changes in motor operating speed shift these components in frequency, reducing the effectiveness of detectors calibrated to a fixed frequency band. We propose a speed-adaptive FPGA-based spectral fault-signature detection pipeline that adjusts its analysis band according to a known operating speed using a precomputed lookup table. The system performs FFT-based spectral analysis and accumulates energy within the speed-dependent frequency band. The complete pipeline is designed for implementation on a Spartan-7 FPGA. Experiments evaluate fixed-band and speed-adaptive detection across multiple operating speeds, signal-to-noise ratios, and fault severities, while also characterizing the FPGA resource and timing cost.

1. Introduction

Induction motors are widely used in industrial systems, where unexpected mechanical or electrical faults can lead to performance degradation and unplanned downtime. Vibration analysis provides a useful means of detecting such faults because different fault conditions can produce characteristic frequency components in the measured vibration signal. For example, rotational-speed-related components and their harmonics can be associated with different mechanical and electrical fault signatures. FPGA-based signal-processing systems have therefore been investigated for performing vibration analysis and fault detection in real time.

FPGA implementations are particularly attractive for this application because signal-processing operations such as filtering, spectral analysis, and feature extraction can be performed using dedicated hardware with predictable timing. Shome *et al*. [9] presented an FPGA-based signal prefiltering system for vibration analysis of induction motor failures, while Contreras-Medina *et al*. [4] developed an FPGA-based vibration analyser incorporating FFT-based analysis and threshold-based detection. These works demonstrate the feasibility of implementing vibration-analysis pipelines on FPGA platforms. However, in the approaches considered in this work, the monitored frequency region is associated with a fixed operating speed, leaving the effect of operating-speed variation on the selected detection band unaddressed.

In practical motor-drive systems, the operating speed may vary over time. Since several fault signatures are related to rotational speed, a change in RPM can shift their corresponding frequency components. Consequently, a narrow-band detector configured for one operating speed may no longer cover the relevant spectral region when the motor operates at a different speed. This creates a need for a detection mechanism that can adapt its analysis band according to the operating speed without introducing excessive hardware complexity.

In this work, we propose a speed-adaptive FPGA-based spectral fault-signature detection pipeline implemented on a Spartan-7 FPGA. The proposed system assumes that the operating RPM is available as an input to the detection logic. Rather than performing a hardware division to calculate the FFT-bin location of the expected fault signature, a small speed-indexed lookup table is precomputed and stored in the FPGA. The lookup table provides the corresponding FFT-bin range for each tested operating speed, allowing the detector to shift its narrow-band energy accumulator as the speed changes.

The proposed pipeline consists of signal generation, FFT-based spectral analysis, Bartlett periodogram calculation, magnitude/power calculation, and threshold-based fault detection. A Python floating-point model is used as a reference for signal generation, lookup-table construction, fixed-point analysis, and verification of the FPGA implementation. The system is evaluated using multiple operating speeds, signal-to-noise ratios, and fault severities.

2. Problem Statement

Can a resource-efficient, speed-adaptive narrowband vibration fault detector implemented on an FPGA maintain reliable fault detection in induction motors under variable-speed operating conditions, and how do variations in mo-

tor speed, fault severity, and measurement noise affect its detection accuracy, response time, and hardware resource utilisation?

3. Literature Review

Contreras-Medina *et al*. [4] investigated an FPGA-based, multichannel vibration analysis system for detecting induction motor faults. The system uses accelerometer measurements to acquire motor vibration signals, which are digitized using an analog-to- digital converter (ADC). The sampled signals are processed using the Fast Fourier Transform (FFT) to obtain their frequency spectra. Fault-specific frequency components are then analyzed to calculate indicators that distinguish healthy and faulty motor conditions.

The study employed a three-axis accelerometer because fault-induced vibrations are not equally observable along all three axes. Experimental observations indicated that broken rotor bar faults are most prominent along the Y-axis, unbalance along the Z-axis, and mechanical looseness along the X-axis. This supports the use of multichannel vibration analysis for capturing fault information across different directions.

The paper mentions that MCSA (motor current signature analysis) is another popular technique to measure motor failure, which gives good results, but it requires dedicated instrumentation and software for each failure, so they used vibration analysis. Motor Current Signature Analysis (MCSA) works by monitoring the motor's stator current and matching the signal to a specific motor fault. This technique mainly uses two approaches: 1) steady-state analysis and 2) startup transient analysis.

In steady-state analysis, Ayhan *et al*. [1] focused on detecting broken rotor bars by looking at the sideband frequencies in the current spectrum when the motor is damaged. Steady-state current monitoring can also detect other types of failures like bearing damage, air-gap eccentricity, and load problems, as described by Benbouzid [2]. Additionally, Ponci *et al*. [7] used steady-state analysis with the Discrete Wavelet Transform (DWT) to detect electrical failures like short circuits or open phases. Finally, startup transient analysis can also be used to detect motor faults, as demonstrated by Ordaz-Moreno *et al*. [6].

4. Proposed Signal Processing Workflow

4.1. Signal Processing and FFT on FPGA

Initially, the vibration signal is in the time domain, and its Discrete Fourier Transform (DFT) is computed using the FFT algorithm:

$$X(k) = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} \tag{1}$$

where $x[n]$ is the input vibration signal, N is the block size, k is the frequency bin index, and $X(k)$ is the complex frequency-domain output.

The reference paper used a radix-2 butterfly algorithm. A radix-2 butterfly takes 2 inputs and produces 2 outputs, reducing the computational complexity from $\mathcal{O}(N^2)$ down to $\mathcal{O}(N \log_2 N)$. The system performs FFT processing for all three axes independently (channel-level parallelism). The paper reports three FFT units and three periodogram estimators in its FPGA implementation, taking 1.33 ms to compute the FFT. Alternatively, Vite- Frias *et al*. [11] implemented a 1024-point FFT using a radix-4 butterfly structure, which further reduces processing latency at the cost of a more complex hardware layout.

Table 1. FPGA Implementation Parameters

Parameter	Implementation
FPGA	Xilinx Spartan-3
FPGA clock	25 MHz
Vibration sensor	Three-axis accelerometer
Input channels	X, Y, Z
Sampling frequency	1500 Hz
FFT size	1024 points
Frequency resolution	Approximately 1.46 Hz
Nyquist frequency	750 Hz
Bartlett averaging	Configurable, 1–32 periodograms

4.2. Bartlett Periodogram

To minimize variance and improve spectrum estimation reliability under noisy conditions, the Bartlett averaged periodogram method is used [8]:

$$P_{xx}^{(i)}(k) = \frac{|X_i(k)|^2}{N} \tag{2}$$

$$P_{xx}^{B}(k) = \frac{1}{M} \sum_{i=0}^{M-1} P_{xx}^{(i)}(k) \tag{3}$$

where M is the number of signal segments averaged. Configuring M from 1 to 32 allows a trade-off between estimate stability and computation time.

4.3. Post-Processing and Feature Extraction

After obtaining the spectrum, the system selects a narrow frequency band where a fault signature is expected. Table 2 lists established relationships between mechanical and electrical failures and their characteristic frequencies [3, 5, 10]. A weighted energy accumulation of the spectral power within the selected frequency range is used to calculate the fault indicator A:

Table 2. Frequency Components and Possible Motor Failures

Frequency	Possible Failure
$1 \times$ RPM	Unbalance, looseness, misalignment of gear or pulley, resonance, electrical problems, alternative forces
$2 \times$ RPM	Mechanical offset, misalignment, alternative forces, resonance
$3 \times$ RPM	Misalignment, axial mechanical beat
$< 1 \times$ RPM	Slip, oil whirl
Supply frequency	Electrical problems
RPM harmonics	Broken bars, damaged gears, aerodynamic forces, hydraulic forces, mechanical set, alternative forces

$$A = \sum_{k=L_I}^{L_S} a_j P_{xx}^B(k) \tag{4}$$

where L_I and L_S are lower and upper bin limits, and a_j is a calibrated weighting factor.

4.4. FPGA Implementation

The reviewed paper's system floor plan is as follows

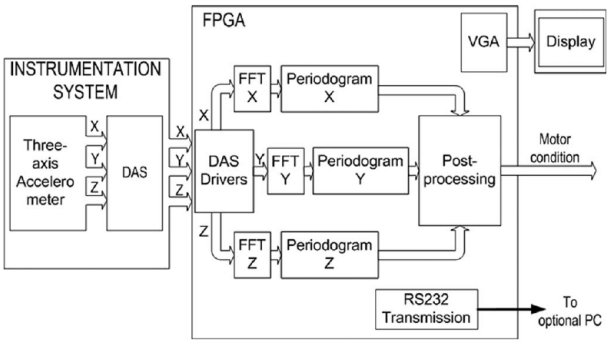


Figure 1. System Floor Plan

4.4.1 Peripherals

VGA Interface: Used to display vibration-analysis results and system information visually.

RS-232 serial interface: Provides optional serialcommunication for transmitting information between the FPGA system and an external device, such as a computer.

Three-axis accelerometer interface: Acquires vibration signals along the X, Y and Z axes for processing by the FPGA.

5. Proposed Speed-Adaptive Methodology

Unlike the reference paper [4], which monitors faults at a single fixed motor speed, our proposed pipeline adapts to motor operating speed variations. The system evaluates speed-adaptive band shifting across three discrete test speeds (nominal, $+20\%$, -20%) using a precomputed lookup table, avoiding the need for an expensive hardware divider.

5.1. Signal Generation

Synthetic three-channel (X, Y, Z) vibration signals are generated in Python to simulate an induction motor under healthy and faulty conditions. The operating RPM is provided as an input signal to the FPGA logic

5.2. Radix-4 FFT Spectral Analysis

Instead of the radix-2 FFT used in the reference paper, a 1024-point radix-4 FFT architecture is implemented. This reduces calculation latency and increases throughput on the Spartan-7 FPGA.

5.3. Bartlett Periodogram

The Bartlett averaged periodogram is computed to estimate the power spectral density across the three vibration channels, reducing spectral variance.

5.4. Speed-Adaptive Lookup Table (LUT) Post-Processing

Frequency bin ranges $[L_I, L_S]$ corresponding to target fault types are selected dynamically using a small speed-indexed lookup table (LUT) based on the input RPM. The weighted fault feature A is accumulated over the shifted bin range.

5.5. Threshold-Based Fault Decision

The calculated feature A is compared against a pre-calibrated threshold τ. If $A > \tau$, a fault flag is generated, identifying the specific fault type.

6. Future Work

Upon successful validation of the proposed system, the following extensions will be explored in the given order:

1. **Additional Fault Signature Support:** Extend the lookup table and feature extractor to support broader fault types (e.g., bearing outer-race defects).

2. **Hann Windowing Integration:** Evaluate the impact of pre-FFT Hann windowing on reducing spectral leakage and improving weak fault detection.

3. **Periodic Anomaly Detection:** Introduce an additional postprocessing mode that periodically performs broader spectral analysis to identify potential anomalies outside the monitored frequency bands.

4. **Machine Learning Classification:** If time permits, try machine learning classification using the University of Ottawa dataset. Since this dataset contains only single-axis vibration measurements, its applicability to our three-axis system will also be investigated.

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